

Multi-Cloud Threat Detection Lab — Technical Report

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Executive Summary

This report documents the design, implementation, and findings of a production-grade multi-cloud threat detection lab built across Amazon Web Services (AWS) and Microsoft Azure. The lab was built to simulate real-world adversary behavior and validate detection capabilities using a centralized Elastic SIEM. All infrastructure was provisioned as code using Terraform, demonstrating a disciplined approach to repeatable and auditable security environments.

The lab successfully ingested **6,466 CloudTrail events** from AWS into Elastic SIEM via a fully automated CloudTrail → S3 → SQS → Elastic Agent pipeline. Four custom KQL detection rules were developed and mapped to the MITRE ATT&CK framework, and a **live High severity alert (Risk Score 73)** was triggered on an IAM credential creation event — validating end-to-end detection capability.

Objectives

The primary objectives of this lab were to:

- Build a multi-cloud log ingestion pipeline from AWS and Azure into a centralized Elastic SIEM
 - Provision all infrastructure using Terraform to demonstrate infrastructure-as-code (IaC) proficiency
 - Develop custom KQL detection rules mapped to real-world MITRE ATT&CK techniques
 - Validate detection logic by triggering and capturing live security alerts
 - Simulate the kind of threat detection environment found in enterprise SOC and cloud security engineering roles
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Threat Model

The lab was designed around a core adversary scenario: an attacker who has gained initial access to an AWS environment and is attempting to establish persistence, escalate privileges, and evade detection.

The following threat scenarios were modeled:

Scenario	MITRE Tactic	Technique
Attacker creates IAM access key to maintain persistent access	Persistence	T1098
Attacker makes unauthorized API calls that return AccessDenied errors, indicating enumeration or privilege testing	Defense Evasion	T1078
Attacker or insider logs into the AWS root account console	Privilege Escalation	T1078.004
Attacker modifies security group rules to open inbound access	Defense Evasion	T1562.007

These scenarios were chosen because they represent realistic early-stage cloud intrusion behaviors commonly observed in cloud breach reports and adversary simulations. They are also directly observable through CloudTrail API logs, making them ideal candidates for KQL-based detection in Elastic SIEM.

Architecture & Infrastructure

Overview

The lab consists of two cloud environments feeding into a single Elastic SIEM instance. All resources were provisioned using Terraform, ensuring reproducibility and eliminating manual configuration drift.

AWS Pipeline

IAM / EC2 API Calls → CloudTrail (logging) → S3 Bucket (storage) → SQS Queue → Elastic Agent → Elastic SIEM

Components provisioned via Terraform:

- **VPC + Subnet** — Isolated network (`multi-cloud-security-lab`, CIDR 10.0.0.0/16) to contain lab resources
- **S3 Bucket** — `security-lab-cloudtrail-logs-769e08c5` receives all CloudTrail log files
- **CloudTrail Trail** — `security-lab-trail` configured as a multi-region trail, actively logging all management and data events
- **SQS Queue** — `cloudtrail-logs-queue` receives S3 event notifications when new log files are delivered, triggering the Elastic Agent to pull and ingest them
- **IAM User + Policy** — Least-privilege IAM user created for the Elastic Agent with permissions scoped to S3 read and SQS consume only

Azure Pipeline

Azure AD / Resource Activity → Azure Monitor → Event Hub → Elastic Agent → Elastic SIEM

Components provisioned via Terraform:

- **Resource Group** — `multi-cloud-security-lab` in East US
- **Virtual Network + Subnet** — `security-lab-vnet` for network isolation
- **Event Hub Namespace + Event Hub** — `security-lab-eventhub` with `insights-operational-logs` hub configured for log streaming
- **Storage Account** — `seclabelasticsur3jj` supporting Event Hub checkpoint storage
- **Log Analytics Workspace** — `security-lab-workspace` for Azure-native log aggregation
- **Microsoft Defender for Cloud** — Enabled for threat detection at the Azure resource level
- **Azure Monitor Diagnostic Settings** — Configured to route Azure Activity Logs to the Event Hub for Elastic ingestion

Elastic SIEM

- **Elastic Agent** running on `Security-Lab-Policy` with both `aws-security-lab` (S3/SQS) and `azure-1` integrations active
- **Fleet** managing agent health and policy enforcement
- **Custom detection rules** running on a 5-minute schedule

Detection Logic

Four custom KQL detection rules were developed in Elastic SIEM, each mapped to a specific MITRE ATT&CK technique. All rules query the `logs-aws.cloudtrail-*` index.

Rule 1 — AWS IAM Access Key Created by IAM User

- **Tactic:** Persistence
- **Technique:** T1098 — Account Manipulation
- **Severity:** High | **Risk Score:** 73
- **Rationale:** Attackers who gain access to an AWS environment frequently create new IAM access keys to maintain persistent programmatic access even if the original credentials are rotated. This rule detects `CreateAccessKey` API calls made by IAM users, which should be rare in a well-managed environment and warrants immediate investigation.

Rule 2 — AWS Unauthorized API Call — AccessDenied

- **Tactic:** Defense Evasion
- **Technique:** T1078 — Valid Accounts
- **Severity:** Medium | **Risk Score:** 47
- **Rationale:** A high volume of `AccessDenied` errors from a single identity is a strong indicator of privilege enumeration or an attacker testing the boundaries of a compromised account. Legitimate users rarely trigger sustained `AccessDenied` responses. This rule surfaces that pattern for analyst review.

Rule 3 — AWS Root Account Console Login Detected

- **Tactic:** Privilege Escalation
- **Technique:** T1078.004 — Cloud Accounts
- **Severity:** High | **Risk Score:** 73
- **Rationale:** AWS best practices strictly prohibit routine use of the root account. Any console login to the root account — whether by an insider or an attacker who has obtained root credentials — is a critical security event that demands immediate response. CloudTrail logs `ConsoleLogin` events that identify root account usage.

Rule 4 — AWS Security Group Modified

- **Tactic:** Defense Evasion
- **Technique:** T1562.007 — Disable or Modify Cloud Firewall
- **Severity:** Medium | **Risk Score:** 47

- **Rationale:** Security group modifications are a common attacker technique to open inbound ports and establish persistent access or lateral movement paths into cloud infrastructure. This rule monitors for `AuthorizeSecurityGroupIngress` and related API calls that indicate firewall rule changes.

Alert Analysis

Live Alert: AWS IAM Access Key Created by IAM User

During lab validation, the IAM access key detection rule successfully fired a **High severity alert** with the following details:

Field	Value
Timestamp	Jun 29, 2026 @ 19:52:06.859
Rule	AWS IAM Access Key Created by IAM User
Severity	High
Risk Score	73
Event Category	IAM
Source IP	73.237.86.152
Reason	IAM event with source IP created high alert — AWS IAM Access Key creation detected

Analysis:

The alert was triggered by a `CreateAccessKey` API call captured in CloudTrail and ingested through the S3 → SQS → Elastic pipeline. The event was categorized under the IAM event category and correctly identified as a Persistence technique under MITRE ATT&CK T1098. The rule executed successfully on its scheduled interval and generated an open alert visible in the Elastic Security Alerts view.

This confirmed that the full detection pipeline — from AWS API call to CloudTrail log to S3 delivery to SQS notification to Elastic ingestion to KQL rule evaluation to alert generation — is functioning end to end.

Gaps & Limitations

An honest assessment of current lab limitations:

- **Azure pipeline not yet validated end to end** — The Event Hub and Diagnostic Settings are provisioned and active, but ingestion of Azure Activity Logs into Elastic SIEM has not been fully confirmed. This remains an open item.
- **No simulated attacker tooling** — Threat scenarios were triggered manually via the AWS console rather than through adversary simulation tools such as Atomic Red Team or Stratus Red Team. Adding automated attack simulation would increase detection coverage confidence.
- **Limited log volume on Azure side** — Without confirmed Azure log ingestion, the multi-cloud detection claim is currently AWS-validated only.
- **No automated response actions** — Detection rules fire alerts but do not trigger any automated response such as isolating a resource, revoking credentials, or notifying via a ticketing system. This would be a natural next step toward a production SOC workflow.
- **Single agent deployment** — The Elastic Agent runs on a single MacBook Pro host rather than a cloud-hosted endpoint, which differs from a production deployment architecture.

Future Improvements

- **Complete and validate Azure Activity Log → Event Hub → Elastic pipeline** to achieve true multi-cloud detection
 - **Integrate Atomic Red Team or Stratus Red Team** for automated adversary simulation against both cloud environments
 - **Add automated response actions** in Elastic — for example, triggering a webhook to a Slack channel or a Lambda function to revoke an IAM key when the persistence rule fires
 - **Expand detection rule coverage** to include additional MITRE ATT&CK techniques such as S3 bucket policy modifications (T1530), CloudTrail logging disabled (T1562.008), and new IAM user creation (T1136.003)
 - **Add Azure-specific detection rules** targeting Azure AD sign-in anomalies and resource group modifications once the Azure pipeline is validated
 - **Migrate Elastic Agent to a cloud-hosted EC2 or Azure VM** to better reflect production deployment architecture
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Conclusion

This lab demonstrates practical, hands-on proficiency across cloud security engineering, infrastructure as code, log pipeline architecture, SIEM detection engineering, and MITRE ATT&CK-mapped threat modeling. The AWS detection pipeline is fully operational, with 6,466 CloudTrail events ingested and a live High severity alert validated. The Azure infrastructure is provisioned and ready for pipeline validation.

The skills demonstrated here — Terraform-provisioned multi-cloud infrastructure, custom KQL detection logic, and end-to-end alert validation — directly map to the responsibilities of cloud security engineer, detection engineer, and SOC analyst roles in enterprise environments.

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